

Pregunta 1

$$I = \int \frac{(x^2 - 1)dx}{(x^4 + 3x^2 + 1) \arctan\left(x + \frac{1}{x}\right)}$$

Dividiendo el numerador y el denominador entre x^2 :

$$I = \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\left(x^2 + 3 + \frac{1}{x^2}\right) \arctan\left(x + \frac{1}{x}\right)} = \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\left[\left(x + \frac{1}{x}\right)^2 + 1\right] \arctan\left(x + \frac{1}{x}\right)}$$

$$\text{Sea } u = x + \frac{1}{x} \implies du = \left(1 - \frac{1}{x^2}\right) dx.$$

$$I = \int \frac{du}{(u^2 + 1) \arctan(u)}$$

$$\text{Sea } v = \arctan(u) \implies dv = \frac{du}{u^2 + 1}.$$

$$I = \int \frac{dv}{v} = \ln |v| + C = \ln |\arctan(u)| + C$$

Retornando a la variable x :

Respuesta:

$$I = \ln \left| \arctan \left(x + \frac{1}{x} \right) \right| + C$$

Pregunta 2

$$I = \int \frac{(\operatorname{sen}(x) + x \cos(x))dx}{(1 - x \operatorname{sen}(x))\sqrt{x^2 - 1 - x^2 \cos^2(x)}}$$

Acomodamos la expresión dentro de la raíz:

$$x^2 - 1 - x^2 \cos^2(x) = x^2(1 - \cos^2(x)) - 1 = x^2 \operatorname{sen}^2(x) - 1$$

Sustituimos $u = x \operatorname{sen}(x) \implies du = (\operatorname{sen}(x) + x \cos(x))dx$:

$$I = \int \frac{du}{(1 - u)\sqrt{u^2 - 1}} = - \int \frac{du}{(u - 1)\sqrt{u^2 - 1}}$$

$$\text{Sea } u - 1 = \frac{1}{v} \implies u = \frac{1}{v} + 1 \implies du = -\frac{1}{v^2}dv$$

$$\text{Para la raíz: } \sqrt{u^2 - 1} = \sqrt{\left(\frac{1}{v} + 1\right)^2 - 1} = \sqrt{\frac{1}{v^2} + \frac{2}{v}} = \frac{\sqrt{1 + 2v}}{v}$$

$$I = - \int \frac{-\frac{1}{v^2}dv}{\left(\frac{1}{v}\right)\left(\frac{\sqrt{1+2v}}{v}\right)} = \int \frac{dv}{\sqrt{1+2v}}$$

$$I = \sqrt{1+2v} + C$$

Retornando a la variable u , con $v = \frac{1}{u-1}$:

$$I = \sqrt{1 + \frac{2}{u-1}} + C = \sqrt{\frac{u+1}{u-1}} + C$$

Retornando a la variable x :

Respuesta:

$$I = \sqrt{\frac{x \operatorname{sen}(x) + 1}{x \operatorname{sen}(x) - 1}} + C$$

Pregunta 3

$$I = \int \frac{4 \operatorname{sen}(2x) + 8 \operatorname{sen}(x) + \cos(x)}{3 \operatorname{sen}(2x) + 3 \cos(x) + 8 \operatorname{sen}(x) + 4} dx$$

Reescribiendo usando $\operatorname{sen}(2x) = 2 \operatorname{sen}(x) \cos(x)$:

Numerador: $N = 8 \operatorname{sen}(x) \cos(x) + 8 \operatorname{sen}(x) + \cos(x)$

Denominador: $D = 6 \operatorname{sen}(x) \cos(x) + 3 \cos(x) + 8 \operatorname{sen}(x) + 4 = (3 \cos(x) + 4)(2 \operatorname{sen}(x) + 1)$

Descomponemos en fracciones de la forma:

$$\frac{N}{D} = \frac{A \cos(x)}{3 \cos(x) + 4} + \frac{B \operatorname{sen}(x)}{2 \operatorname{sen}(x) + 1}$$

Multiplicando en aspa y comparando coeficientes, obtenemos $A = 1, B = 2$:

$$\frac{\cos(x)(2 \operatorname{sen}(x) + 1) + 2 \operatorname{sen}(x)(3 \cos(x) + 4)}{(3 \cos(x) + 4)(2 \operatorname{sen}(x) + 1)} = \frac{8 \operatorname{sen}(x) \cos(x) + 8 \operatorname{sen}(x) + \cos(x)}{D}$$

Separamos la integral en dos partes $I = I_1 + I_2$:

$$I_1 = \int \frac{\cos(x)}{3 \cos(x) + 4} dx = \frac{1}{3} \int \left(1 - \frac{4}{3 \cos(x) + 4} \right) dx = \frac{x}{3} - \frac{4}{3} \int \frac{dx}{3 \cos(x) + 4}$$

Por sustitución de Weierstrass $t = \tan(x/2)$:

$$\int \frac{dx}{3 \cos(x) + 4} = \int \frac{\frac{2dt}{1+t^2}}{3 \left(\frac{1-t^2}{1+t^2} \right) + 4} = \int \frac{2dt}{t^2 + 7} = \frac{2}{\sqrt{7}} \arctan \left(\frac{t}{\sqrt{7}} \right)$$

$$\Rightarrow I_1 = \frac{x}{3} - \frac{8}{3\sqrt{7}} \arctan \left(\frac{\tan(x/2)}{\sqrt{7}} \right)$$

$$I_2 = \int \frac{2 \operatorname{sen}(x)}{2 \operatorname{sen}(x) + 1} dx = \int \left(1 - \frac{1}{2 \operatorname{sen}(x) + 1} \right) dx = x - \int \frac{dx}{2 \operatorname{sen}(x) + 1}$$

Por sustitución de Weierstrass $t = \tan(x/2)$:

$$\int \frac{dx}{2 \operatorname{sen}(x) + 1} = \int \frac{\frac{2dt}{1+t^2}}{2 \left(\frac{2t}{1+t^2} \right) + 1} = \int \frac{2dt}{(t+2)^2 - 3} = \frac{1}{\sqrt{3}} \ln \left| \frac{t+2-\sqrt{3}}{t+2+\sqrt{3}} \right|$$

$$\Rightarrow I_2 = x - \frac{1}{\sqrt{3}} \ln \left| \frac{\tan(x/2) + 2 - \sqrt{3}}{\tan(x/2) + 2 + \sqrt{3}} \right|$$

Respuesta:

$$I = \frac{4x}{3} - \frac{8}{3\sqrt{7}} \arctan \left(\frac{\tan(x/2)}{\sqrt{7}} \right) - \frac{1}{\sqrt{3}} \ln \left| \frac{\tan(x/2) + 2 - \sqrt{3}}{\tan(x/2) + 2 + \sqrt{3}} \right| + C$$

Pregunta 4

$$I = \int \frac{\ln(x)dx}{x(\ln^2(x) - 2)\sqrt{\ln^4(x) - 4\ln^2(x) + 5}}$$

Completando cuadrados en el interior de la raíz: $\ln^4(x) - 4\ln^2(x) + 5 = (\ln^2(x) - 2)^2 + 1$

$$\text{Sea } u = \ln^2(x) - 2 \implies du = 2\frac{\ln(x)}{x}dx \implies \frac{\ln(x)dx}{x} = \frac{du}{2}.$$

$$I = \frac{1}{2} \int \frac{du}{u\sqrt{u^2 + 1}}$$

Mediante sustitución trigonométrica $u = \tan(\theta) \implies du = \sec^2(\theta)d\theta$:

$$I = \frac{1}{2} \int \frac{\sec^2(\theta)d\theta}{\tan(\theta)\sec(\theta)} = \frac{1}{2} \int \csc(\theta)d\theta$$

$$I = \frac{1}{2} \ln |\csc(\theta) - \cot(\theta)| + C$$

$$\text{Dado que } \tan(\theta) = u \implies \csc(\theta) = \frac{\sqrt{u^2 + 1}}{u} \text{ y } \cot(\theta) = \frac{1}{u} :$$

$$I = \frac{1}{2} \ln \left| \frac{\sqrt{u^2 + 1} - 1}{u} \right| + C$$

Retornando a la variable x :

Respuesta:

$$I = \frac{1}{2} \ln \left| \frac{\sqrt{\ln^4(x) - 4\ln^2(x) + 5} - 1}{\ln^2(x) - 2} \right| + C$$

Pregunta 5

$$I = \int \frac{(1 - \sqrt{x^2 + x + 1})^3}{x^5 \sqrt{x^2 + x + 1}} dx = \int \left(\frac{1 - \sqrt{x^2 + x + 1}}{x} \right)^3 \frac{dx}{x^2 \sqrt{x^2 + x + 1}}$$

Sea $y = \frac{1 - \sqrt{x^2 + x + 1}}{x}$. Despejando x :

$$xy - 1 = -\sqrt{x^2 + x + 1} \implies x^2 y^2 - 2xy + 1 = x^2 + x + 1 \implies x = \frac{2y + 1}{y^2 - 1}$$

$$\text{Derivando obtenemos: } dx = \frac{-2(y^2 + y + 1)}{(y^2 - 1)^2} dy$$

$$\text{Además: } \sqrt{x^2 + x + 1} = 1 - xy = 1 - \frac{2y^2 + y}{y^2 - 1} = \frac{-y^2 - y - 1}{y^2 - 1}$$

$$\text{Reemplazando en el diferencial compuesto: } \frac{dx}{x^2 \sqrt{x^2 + x + 1}} = \frac{2(y^2 - 1)}{(2y + 1)^2} dy$$

$$I = \int y^3 \frac{2(y^2 - 1)}{(2y + 1)^2} dy = \int \frac{2y^5 - 2y^3}{(2y + 1)^2} dy$$

$$\text{Sea } z = 2y + 1 \implies y = \frac{z - 1}{2} \text{ y } dy = \frac{dz}{2} :$$

$$I = \int \frac{2 \left(\frac{z-1}{2} \right)^5 - 2 \left(\frac{z-1}{2} \right)^3}{z^2} \frac{dz}{2} = \frac{1}{32} \int \frac{(z-1)^5 - 4(z-1)^3}{z^2} dz$$

$$I = \frac{1}{32} \int \left(z^3 - 5z^2 + 6z + 2 - \frac{7}{z} + \frac{3}{z^2} \right) dz$$

$$I = \frac{1}{32} \left(\frac{z^4}{4} - \frac{5z^3}{3} + 3z^2 + 2z - 7 \ln |z| - \frac{3}{z} \right) + C$$

Respuesta:

$$I = \frac{1}{32} \left(\frac{z^4}{4} - \frac{5z^3}{3} + 3z^2 + 2z - 7 \ln |z| - \frac{3}{z} \right) + C, \quad z = 2 \left(\frac{1 - \sqrt{x^2 + x + 1}}{x} \right) + 1$$

Pregunta 6

$$I = \int \frac{x dx}{(x-1)^4 \sqrt{x^2 + 4x + 1}}$$

Aplicamos la sustitución $t = \frac{1}{x-1} \implies x = \frac{t+1}{t}$ y $dx = -\frac{dt}{t^2}$.

Expresamos la raíz en términos de t :

$$\sqrt{x^2 + 4x + 1} = \sqrt{\left(\frac{t+1}{t}\right)^2 + 4\left(\frac{t+1}{t}\right) + 1} = \sqrt{\frac{t^2 + 2t + 1 + 4t^2 + 4t + t^2}{t^2}} = \frac{\sqrt{6t^2 + 6t + 1}}{t}$$

Reemplazamos en la integral original:

$$I = \int \frac{\left(\frac{t+1}{t}\right)}{\left(\frac{1}{t^4}\right) \frac{\sqrt{6t^2+6t+1}}{t}} \left(-\frac{dt}{t^2}\right) = - \int \frac{t^3 + t^2}{\sqrt{6t^2 + 6t + 1}} dt$$

Utilizamos el método de coeficientes indeterminados (Ostrogradsky):

$$\int \frac{t^3 + t^2}{\sqrt{6t^2 + 6t + 1}} dt = (At^2 + Bt + C)\sqrt{6t^2 + 6t + 1} + \lambda \int \frac{dt}{\sqrt{6t^2 + 6t + 1}}$$

Derivando ambos lados con respecto a t :

$$\frac{t^3 + t^2}{\sqrt{6t^2 + 6t + 1}} = (2At + B)\sqrt{6t^2 + 6t + 1} + \frac{(At^2 + Bt + C)(12t + 6)}{2\sqrt{6t^2 + 6t + 1}} + \frac{\lambda}{\sqrt{6t^2 + 6t + 1}}$$

Multiplicando todo por $\sqrt{6t^2 + 6t + 1}$ y agrupando términos:

$$t^3 + t^2 = (2At + B)(6t^2 + 6t + 1) + (At^2 + Bt + C)(6t + 3) + \lambda$$

$$t^3 + t^2 = 18At^3 + (15A + 12B)t^2 + (2A + 9B + 6C)t + (B + 3C + \lambda)$$

Igualando los coeficientes del polinomio:

$$t^3 : \quad 18A = 1 \implies \boxed{A = 1/18}$$

$$t^2 : \quad 15A + 12B = 1 \implies 15(1/18) + 12B = 1 \implies \boxed{B = 1/72}$$

$$t^1 : \quad 2A + 9B + 6C = 0 \implies 2(1/18) + 9(1/72) + 6C = 0 \implies \boxed{C = -17/432}$$

$$t^0 : \quad B + 3C + \lambda = 0 \implies 1/72 + 3(-17/432) + \lambda = 0 \implies \boxed{\lambda = 5/48}$$

Resolvemos la integral restante completando cuadrados:

$$\int \frac{dt}{\sqrt{6t^2 + 6t + 1}} = \frac{1}{\sqrt{6}} \int \frac{dt}{\sqrt{\left(t + \frac{1}{2}\right)^2 - \frac{1}{12}}} = \frac{1}{\sqrt{6}} \ln \left| t + \frac{1}{2} + \frac{\sqrt{6t^2 + 6t + 1}}{\sqrt{6}} \right|$$

Sustituyendo en la ecuación general y multiplicando por el signo negativo inicial:

$$I = - \left[\left(\frac{24t^2 + 6t - 17}{432} \right) \sqrt{6t^2 + 6t + 1} + \frac{5}{48\sqrt{6}} \ln \left| t + \frac{1}{2} + \frac{\sqrt{6t^2 + 6t + 1}}{\sqrt{6}} \right| \right] + C$$

Retornando a la variable x , sabemos que $t = \frac{1}{x-1}$ y $\sqrt{6t^2 + 6t + 1} = \frac{\sqrt{x^2 + 4x + 1}}{x-1}$. Evaluando la parte algebraica:

$$- \left(\frac{24 + 6(x-1) - 17(x-1)^2}{432(x-1)^2} \right) \left(\frac{\sqrt{x^2 + 4x + 1}}{x-1} \right) = \frac{(17x^2 - 40x - 1)\sqrt{x^2 + 4x + 1}}{432(x-1)^3}$$

Respuesta:

$$I = \frac{(17x^2 - 40x - 1)\sqrt{x^2 + 4x + 1}}{432(x-1)^3} - \frac{5}{48\sqrt{6}} \ln \left| \frac{\sqrt{6}(x+1) + 2\sqrt{x^2 + 4x + 1}}{2\sqrt{6}(x-1)} \right| + C$$

Pregunta 7

$$I = \int \frac{\cosh(x)dx}{\sinh(x)\sqrt[3]{1+\sqrt{\sinh(x)}}}$$

Sea $u = \sinh(x) \implies du = \cosh(x)dx$:

$$I = \int \frac{du}{u\sqrt[3]{1+\sqrt{u}}}$$

Sea $v = \sqrt{u} \implies u = v^2$ y $du = 2v dv$:

$$I = \int \frac{2v dv}{v^2\sqrt[3]{1+v}} = 2 \int \frac{dv}{v\sqrt[3]{1+v}}$$

Aplicamos sustitución de Chebyshev, sea $t^3 = 1+v \implies v = t^3 - 1$ y $dv = 3t^2 dt$:

$$I = 2 \int \frac{3t^2 dt}{(t^3 - 1)t} = 6 \int \frac{t dt}{t^3 - 1}$$

Descomponiendo en fracciones parciales:

$$\frac{t}{t^3 - 1} = \frac{1}{3} \left(\frac{1}{t-1} \right) - \frac{1}{3} \left(\frac{t-1}{t^2+t+1} \right)$$

$$I = 2 \int \frac{dt}{t-1} - 2 \int \frac{t-1}{t^2+t+1} dt = 2 \int \frac{dt}{t-1} - \int \frac{2t+1-3}{t^2+t+1} dt$$

$$I = 2 \ln |t-1| - \ln |t^2+t+1| + 3 \int \frac{dt}{(t+\frac{1}{2})^2 + \frac{3}{4}}$$

$$I = \ln \left| \frac{(t-1)^2}{t^2+t+1} \right| + 2\sqrt{3} \arctan \left(\frac{2t+1}{\sqrt{3}} \right) + C$$

Retornando variables con $t = \sqrt[3]{1+\sqrt{\sinh(x)}}$:

Respuesta:

$$I = \ln \left| \frac{(t-1)^2}{t^2+t+1} \right| + 2\sqrt{3} \arctan \left(\frac{2t+1}{\sqrt{3}} \right) + C, \quad t = \sqrt[3]{1+\sqrt{\sinh(x)}}$$